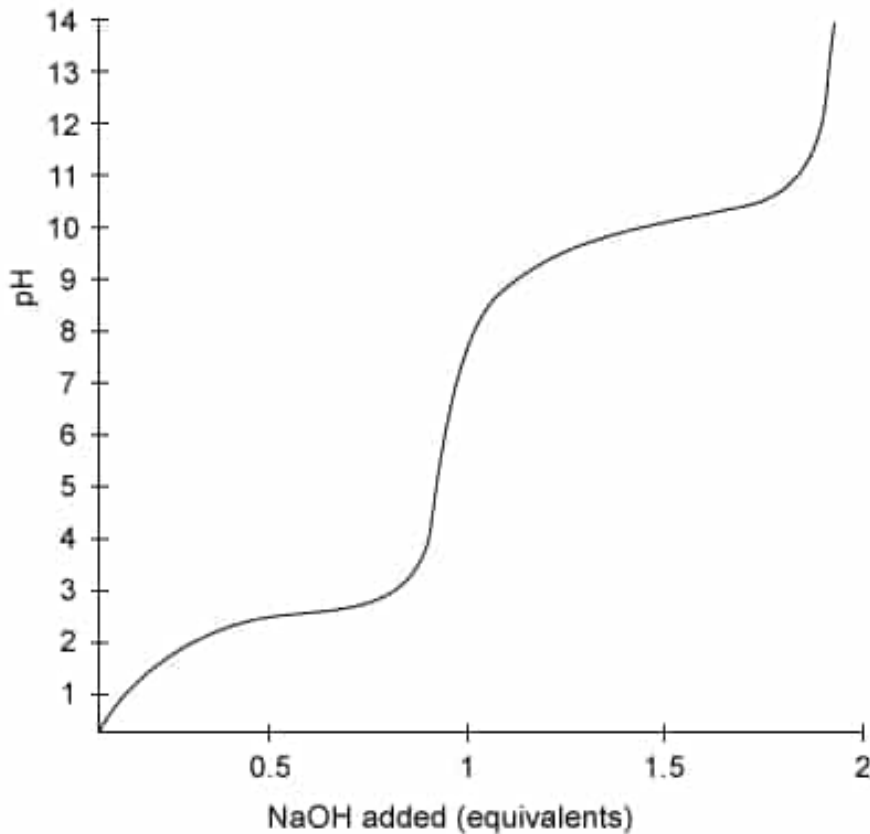


Based on the titration curve of an unknown amino acid, shown below, can the side chain of the amino acid readily exchange protons with water?

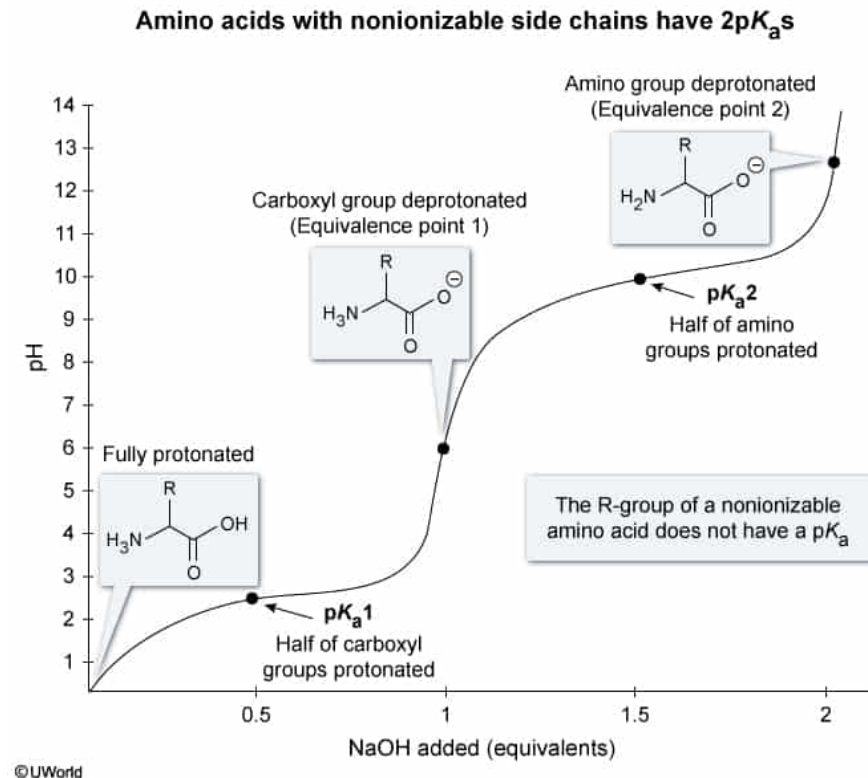


- ☐ A. Yes, because the curve has three equivalence points. [27%]
- ☐ B. Yes, because the curve has only one buffering region. [9%]
- ☐ C. No, because the final pH of the titration is 14. [8%]
- ☒ D. No, because the curve shows only two pK_a values. [54%]

Correct

53% answered correctly

Explanation:



Acidic and basic chemical groups, also known as **ionizable groups**, have intrinsic tendencies to **exchange protons with water**. This tendency is described by the pK_a , or the pH at which half of the ionizable groups of interest are protonated. pK_a values can be determined by titration, which is typically performed by gradually adding a base to an acidic solution of the molecule of interest. As the pH of the solution **approaches the pK_a** of a chemical group, the system acts as a buffer and resists changes in pH by accepting protons from water or donating protons to it. On a **titration curve**, the **region near a pK_a** (called the

buffering region) is **relatively flat** because the addition of base changes the pH by only a small amount.

All amino acids have **ionizable carboxyl and amino groups** with pK_a values near 2 and 9.5, respectively.

Therefore, all amino acid titration curves have **two buffering regions**, one near each pK_a . Seven of the standard amino acids (**R, K, Y, C, H, E, and D**) also have **ionizable side chains**, each with a unique pK_a .

The titration curve for each of these amino acids has a **third buffering region** associated with the side chain pK_a . Because the titration curve in the question shows only two buffering regions, the amino acid has **only two pK_a values**. Therefore, its side chain is not ionizable and **cannot exchange protons** with water.

(Choices A and B) The titration curve corresponds to an amino acid with a non-ionizable side chain, which cannot exchange protons with water. It has only two equivalence points and two buffering regions.

(Choice C) The final pH of a titration is determined by the pH of the base (or acid) that is gradually added during the titration. It is not related to the molecule that is titrated.

Educational objective:

All amino acids have at least two pK_a values that

correspond to their carboxyl and amino groups. Amino acids with ionizable side groups (R, K, Y, C, H, E, and D) have three pK_a values. Titration curves show the region surrounding each pK_a as relatively flat portions of the curve.

Time spent:0 sec

Subject:
Biochemistry

QID:401957

System:
Amino Acids and Proteins